



DBCLS BioHackathon 2026 report: QPX: Pathway analysis environment for non-model organisms

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Introduction

As part of the DBCLS BioHackathon 2026, we here report the development of pathway analysis environment called Quest for Pathways with eXpression (QPX) designed for non-model organisms.

Results and Discussion

Following BH23 (Pico et al., [2023](#)) and BH25 (Bono et al., [2026](#)), discussions continued at BH26 with the aim of enhancing the usability of QPX. At BH26, together with three new members from Hiroshima University (two undergraduate students and one postdoctoral researcher), we worked on the following points toward the development of pathway data for non-model organisms.

PathLift in use

Create new pathways for non-model organisms with public DB ID annotations

QPX in Practice

Transcriptome + Metabolome

Transcriptome + Proteome

QPX AI

Future work

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Author contributions

Hidemasa Bono: Conceptualization, Writing - original draft; Naoya Oec: System development, Validation; Haruka Tanimoto: Data creation; Tomohiro Taguchi: Data creation; Amane Kurata: Data creation; Ryo Nozu: Validation, Data creation.

References

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